From Infrastructure to Impact:
Implementing Sovereign AI at Denvr Dataworks



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Denvr Dataworks

MARKET IMPACT
& ADOPTION

OCP GLOBAL SUMMIT | 2026

Follow Along or Catch Up Later!

Tech Deep Dive Topics:

- **Technical details** Whitepapers, podcasts, and all slides from recent presentations
- **Hardware Innovation** H200 and B300 Immersion Server Development
- **AI Implementation** Challenges and opportunities with a focus on the AI user
- **Open Source Toolkit** Co-optimization across the vertically integrated AI platform
- **Reliability** Multi-year reliability demonstrations – multiple versions including: infrastructure, IT equipment, and fluid

Request a Meeting Schedule a follow up conversation on the same landing page!

Denvr Dataworks

Building Canada's Sovereign AI Cloud

Canadian-built infrastructure. Full-stack AI cloud.
Accelerating what's next in AI.



SOVEREIGN

Canadian owned and operated

SCALABLE

6 month build cycle, 26,000+ GPUs deployed for AI

SERVING

AI Native, Public Sector, Researchers, Enterprise

OCP OPEN COMPUTE PROJECT®

Open standards.
Stronger together.
Building the future.

FOUNDED 2017

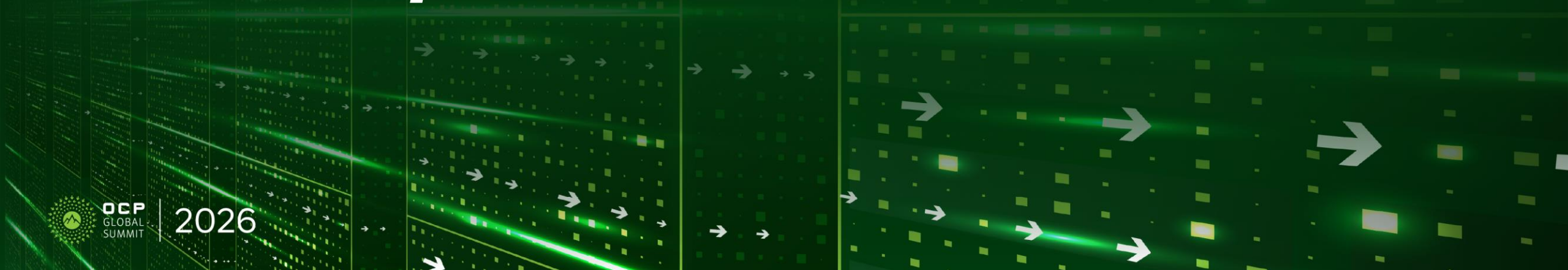
26,000+ GPUS DEPLOYED FOR AI

ECOSYSTEM PARTNERS:

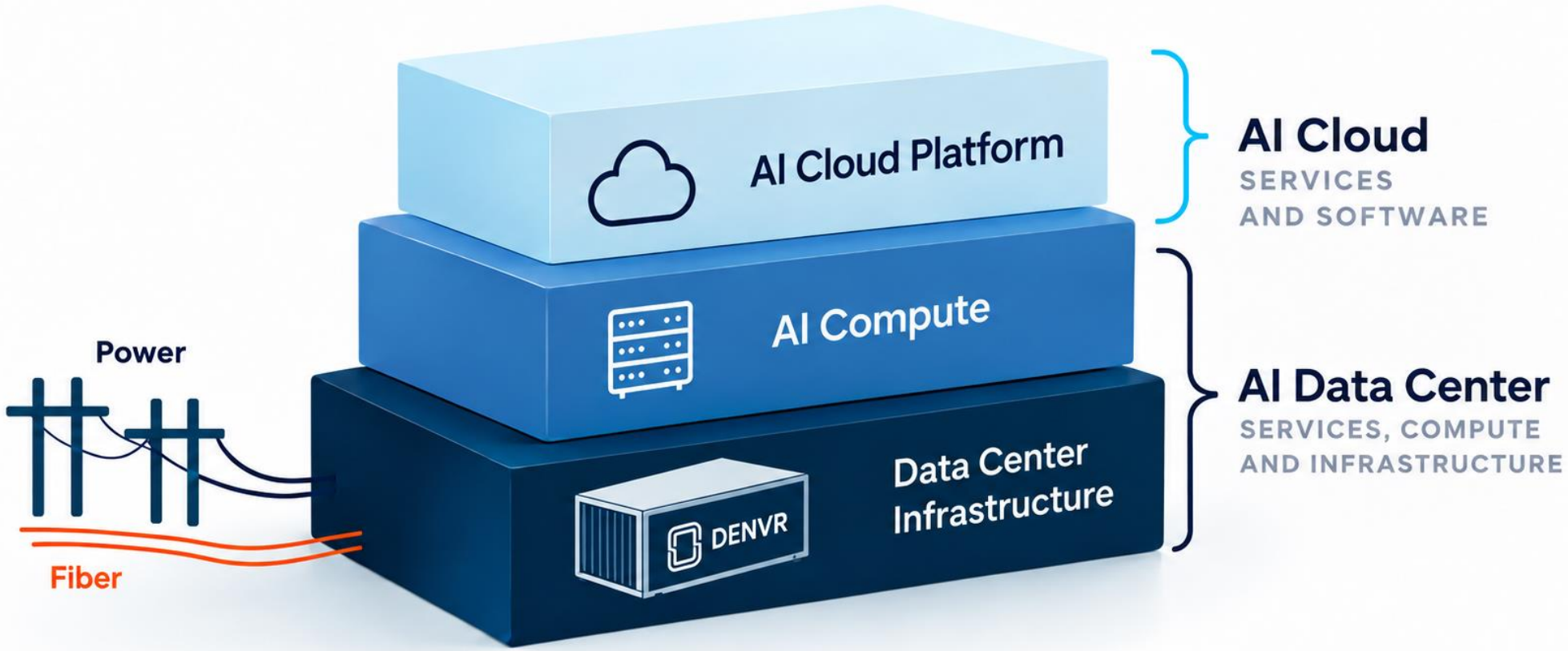
NVIDIA

intel partner

AI impact depends on removing the barriers between an idea and a production outcome.



What Does it Take to Deliver an AI Experience?



Different Customers, Different Barriers

Calgary

1 *Can we control where AI runs, and prioritize sustainability?*

PRECISION AI

2 *Can compute keep pace with our innovation?*

nesi

3 *Can we use AI without exposing proprietary knowledge?*

Same Denvr Platform, Different Valued Features

Calgary

1 *Sovereign, locally connected AI compute for secure and sustainable municipal innovation.*

PRECISION AI

2 *Scalable, on-demand access to dedicated compute, persisted and dedicated storage, & managed networking*

nesi

3 *Dedicated, private AI models and cloud services to unlock proprietary engineering knowledge.*

City of Calgary

A secure, locally connected foundation for municipal AI innovation

Their Denvr Experience:

1. **Private connectivity**
   Sovereignty through infrastructure, connectivity and control

2. **Right-sized compute**
   Local AI capacity for municipal innovation

3. **Zero-water cooling**
   High-density compute; no water consumption







Sovereignty as a Critical Requirement

1	2	3	4	5
Data residency	**Data sovereignty**	**Jurisdictional control**	**Operational sovereignty**	**Full-stack ownership**
Where the bits physically sit. Necessary, but the weakest form of assurance.	Which country's law governs the data itself.	Which government can compel access, regardless of location. Where the CLOUD Act applies.	Who holds the keys, the admin access, and the control plane.	A Canadian-owned company owning and operating the hardware, network and software under Canadian law.

Make it Modular: More AI, Less Footprint

Zero water immersion data centers

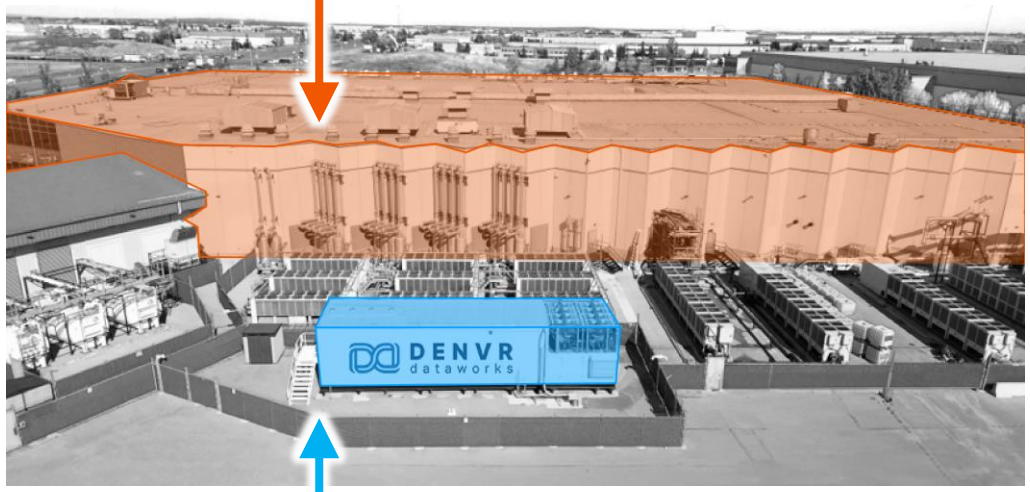


3OU H200 SXM servers deployed 14 per rack at Denvr

~90% smaller footprint;
Security that matches the site

Air Cooled Colocation

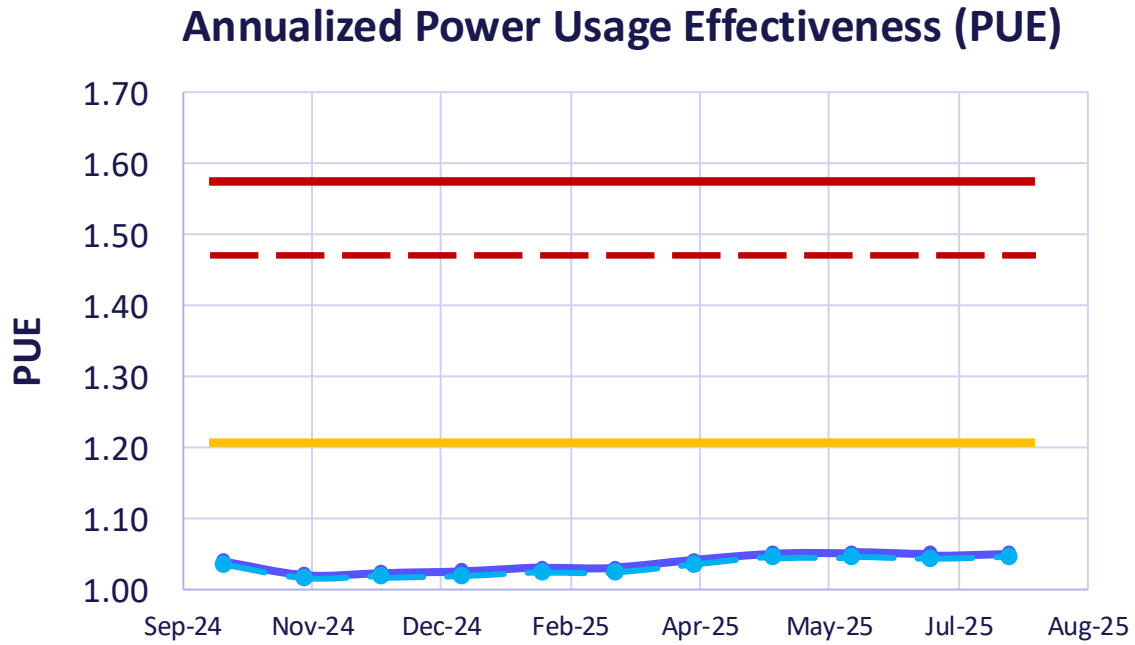
1 MW / 6400 sq. ft.



Denvr Modular Data Center

1 MW / 600 sq ft

Sustainability that Improves with Utilization & TDP



1.58	Global average
1.47	Capacity-weighted global average
1.20	"High efficiency" data center
< 1.04	**Denvr Dataworks**

Best-in-class PUE with power savings of 800 W per server

Precision AI

Scaling AI innovation from the cloud to the field

PRECISION AI

Their Denvr Experience:

1. **AI-Powered Precision Agriculture**
   Accurate identification of individual weeds for targeted crop treatment.

2. **High-Performance AI Compute**
   Enable compute-intensive, bursty and iterative vision model workloads.

3. **Scalable Cloud Infrastructure**
   Flexible, reliable capacity that grows with development & operations.







Designing Infrastructure & ITE for Dynamic Workloads

Optimized & Warrantied Immersion-Ready GPU Servers

- Dell H200 SXM and B300 SXM ORv2.2 and ORv3 servers – learn more about Denvr-Unicom-Dell hardware collab on Wednesday 4:20 PM, room 220C.

Dynamic IT Load

- Validated cooling performance across sustained, intermittent, and idle server workloads for 128-512 GPU server MDCs.

Responsive Cooling

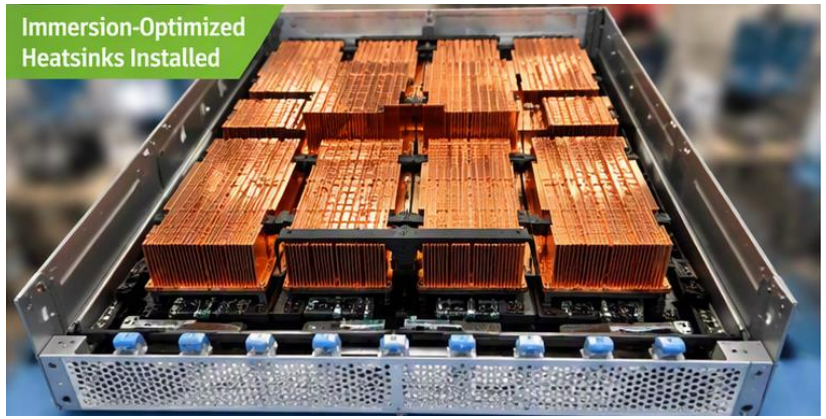
- Fluid flow increases with compute demand and rapidly stabilizes.

Stable Thermal Environment:

- Support bursty and unpredictable workloads



Fourteen H200 SXM servers with 3OU chassis supported in a single Denvr production environment tank.

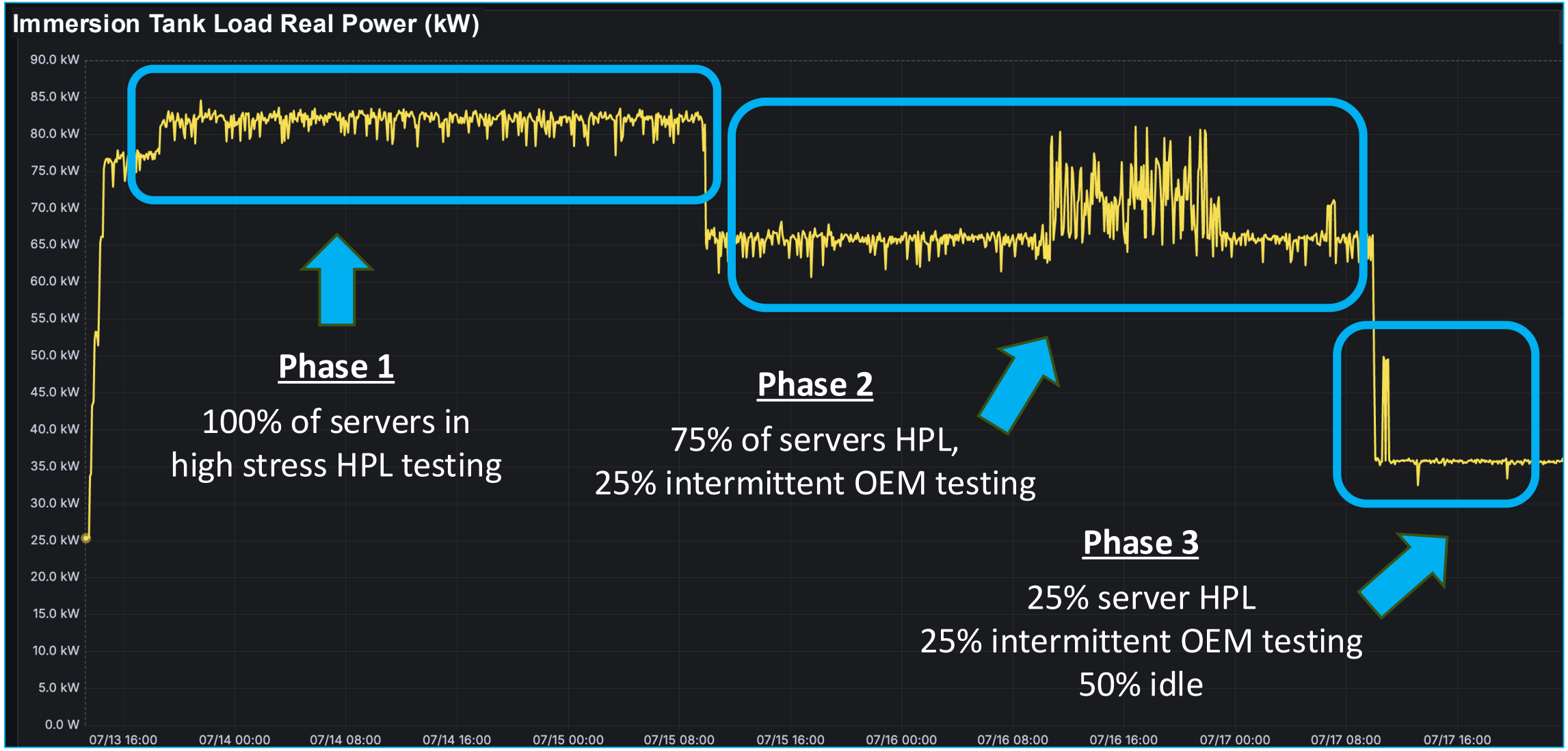


See Denvr-Unicom OCP talk Wednesday to learn about B300 SXM server design and validation work!

Tech deep dive content

OCP GLOBAL SUMMIT | 2026 DENVR UNICOM

Denvr Infrastructure in Dynamic Workloads



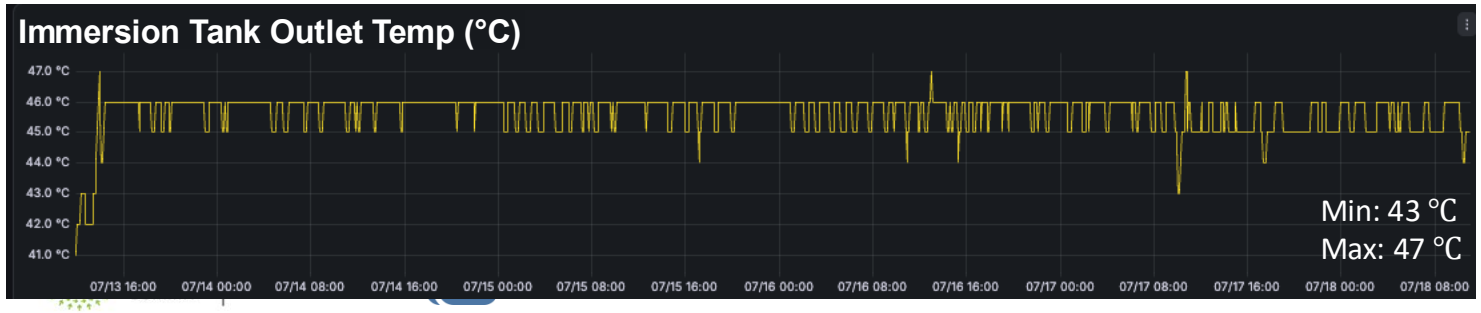
Representative example from Denvr H200 SXM cluster commissioning

Reliability, Responsiveness & Scalability of Denvr Infra









Power Draw

- Measured IT Load responds as expected.
- ~800 W per server power savings vs. air cooled parent configuration.

Flow Rate

- Increases as expected in response to IT load.
- Stabilizes quickly in each test phase.

Mild Temperatures, Longer Lifetime Servers

- Peak system temp increases only 6°C at and quickly stabilizes at ~45 °C.
- Rack inlet/outlet ΔT ~10°C maintained as targeted.
- FMEA testing demonstrated 15 min-1 hr response time possible due to mild thermal env.

Representative example from Denvr H200 SXM cluster commissioning

DENVR

NORAM Electrolysis Systems Inc. (NESI)

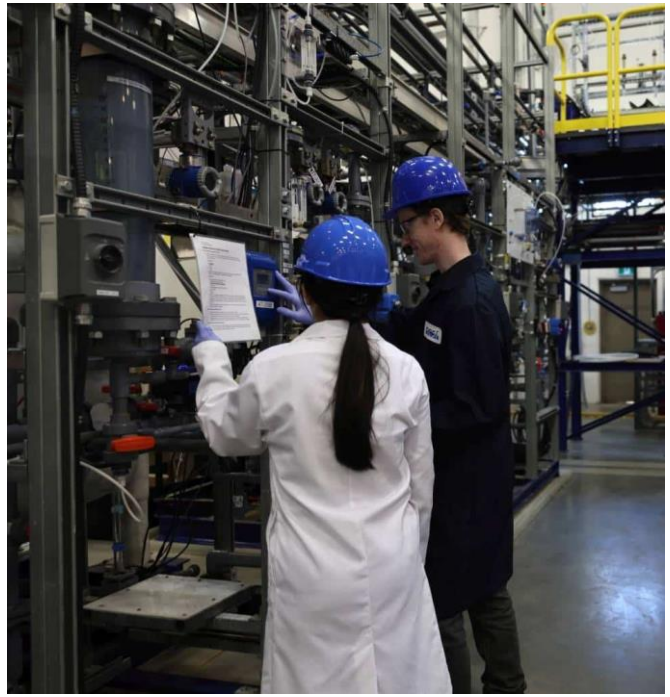
Unlocking years of engineering knowledge with private AI

Their Denvr Experience:

1. **Valorization of Historic Engineering Knowledge**
   Better leverage of proprietary designs, documents and technical expertise.

2. **Security and Control**
   Sensitive intellectual property requires a private AI environment.

3. **Accessible Enterprise AI**
   A searchable knowledge base powered by dedicated AI models and user-selected technical complexity.

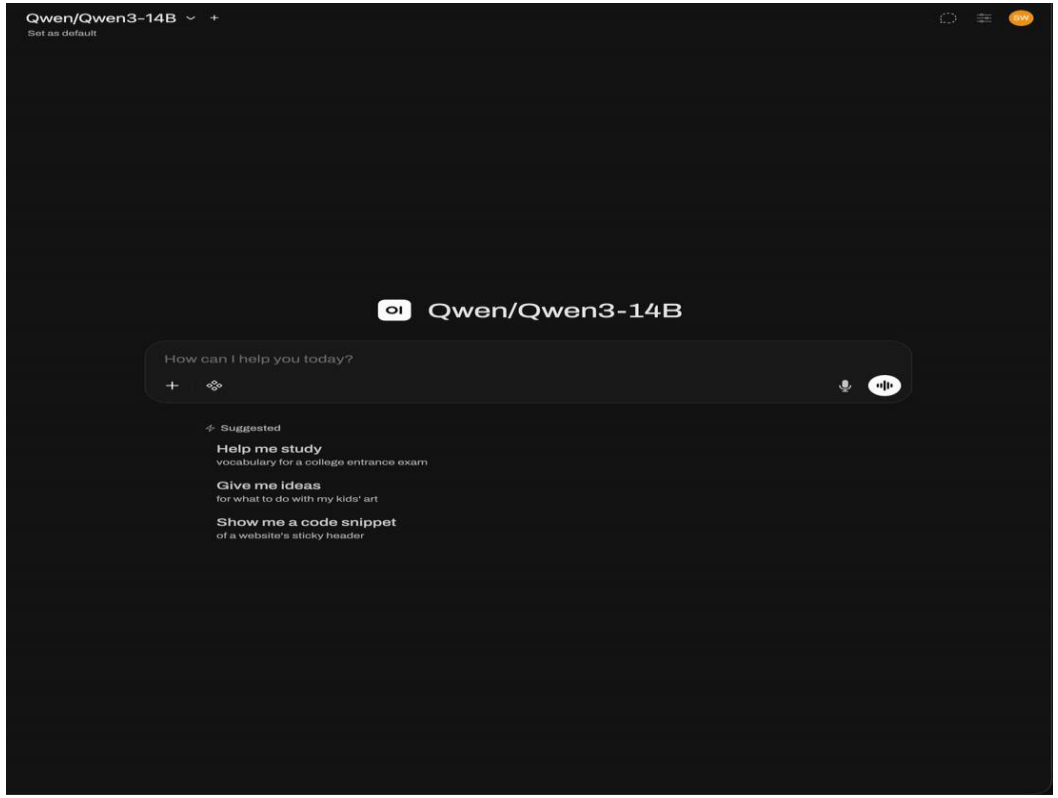




Dedicated Models With Custom Interface

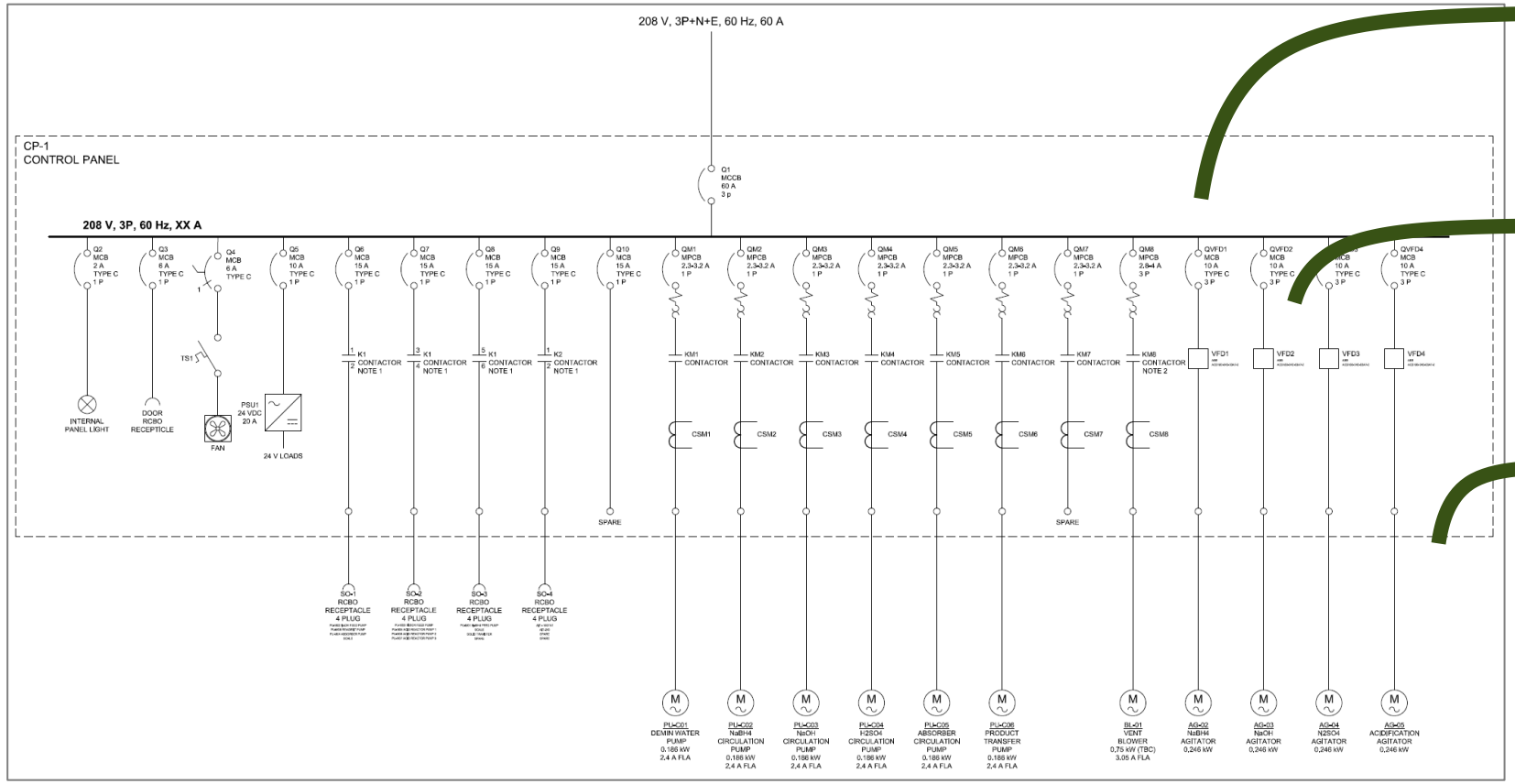
Same Time to Output, Same Model, Different User Experience





- Fully managed and dedicated model application (guaranteed performance and model context)
- OpenAI API compatible endpoints, enables a broad spectrum of use-cases (left)
- Denvr hosted OpenWebUI applications backed by dedicated models (right), leveraging the same endpoints

Establishing a Secure Knowledge Base



Q: Which pumps use VFDs?

A: AG-02, AG-03, AG-04, AG-05

Q: What size DC power supply is used for this design?

A: 20 amps

Q: What is the TCO for this design vs. the alternative?

A: This design enables an additional 10% savings; Data roll up shows an itemized list of total components and associated costs

Simplifying the process of working with unstructured, heterogeneous data

OCP Is Accelerating Our Journey – Please Join In!!

OCP Leadership:

- **Cooling Environments Technical Committee Chair**
  Immersion Solutions
- **Workstream Leadership**
  AI Neocloud
  Immersion Requirements

Denvr Dataworks Team Contributions:

- Open Systems for AI – Blueprint for Scalable Infrastructure
- ORv3 Power Distribution Base Specification
- Immersion Requirements
- Hardware Warranty
- Server Component Immersion Compatibility Testing
- Safe Handling Guidelines
- Ignition Risk Assessment and Compliance Landscape (*forthcoming*)



OCP Cooling Environments Leadership Team



Denvr's scalable deployment module

Thank You!

Access these slides and other technical deep dive content.

Schedule a meeting with our team.

After-hours discussion with Denvr – Ask Me Anything!

